

C++

PENG

Created: 20260817, Version: 20260831

Contents

1	Variables and Basic Types	1
1.1	Primitive Built-in Types	1
1.1.1	Arithmetic Types	1
1.1.2	Type Conversions	1
1.1.3	Literals	2
1.2	Variables	2
1.2.1	Variable Definitions	2
1.2.2	Variable Declarations	2
1.2.3	Identifiers	2
1.2.4	Scope of a Name	2
1.3	Compound Types	3
1.3.1	References	3
1.3.2	Pointers	3
1.4	const	3

1 Variables and Basic Types

1.1 Primitive Built-in Types

1.1.1 Arithmetic Types

Definition 1.1 (Primitive builtin types [1]).

- *arithmetic types*:
 - *integral types*
 - * *characters*: `char`, `std::string`
 - * *integers*: `int`: *signed or unsigned* (≥ 0)
 - * *boolean values*: `bool`
 - *floating-point types*: `double`
- *void*

Remark 1.2. Use `double` for *floating-point computations*.

1.1.2 Type Conversions

Example 1.3 (Type conversions).

```
bool b = 20; // 0 for false and nonzero values for true
int i = b;   // true for 1 and false for 0
```

Example 1.4 (Signed and unsigned).

```
unsigned u = -1; // if 32-bit ints, 4294967295
int a = -1;
unsigned b = 1;
a * b // int value is converted to unsigned before addition
```

1.1.3 Literals

Definition 1.5 (Literals).

Integer literals:

- *decimal*
- *octal*: 0-
- *hexadecimal*: 0x-
- *suffix*:
 - *u*: *unsigned*
 - *L*: *long*

Floating-point literals (double by default):

- *decimal point*
- *E*: 1E-2
- *suffix*: *f*/*F*: *float*

character and string literals: "Hello" = "Hello\0"

Definition 1.6 (Escape sequences).

- *newline*: \n
- *horizontal tab*: \t
- *null*: \0

1.2 Variables

1.2.1 Variable Definitions

Remark 1.7.

- *Variables defined outside any function body are initialized to 0.*
- *Initialize every object of built-in type.*
- `std::cin >> int i;` *error, should be* `int i; std::cin >> i;`.

1.2.2 Variable Declarations

- File A define the variable `int ix;`. Other files: `extern int ix;`.
- All variables must be declared before being used.

1.2.3 Identifiers

Naming conventions:

- Variable name: lowercase
- Class name: uppercase

1.2.4 Scope of a Name

- Global scope: `::global_var`
- Block scope

Remark 1.8. *It is almost always a bad idea to define a local variable with the same name as a global variable that the function uses.*

1.3 Compound Types

1.3.1 References

Definition 1.9 (Reference). *A reference defines an alternative name for an object.*

```
int ival = 0;
int &refVal = ival;
```

- A reference is an alias, not an object.
- The type of a reference and the object must match exactly.
- A reference can only refer to the object to which it was initially bound.

1.3.2 Pointers

Definition 1.10 (Pointer). *A pointer holds the address of another object.*

```
int ival = 0;
int *pi = &ival;
*pi = 10;
int* p1, p2; // legal, but might be misleading. p2 is int.
```

- A pointer is an object. The type of the pointer and the object to which it points must match.
- Dereferencing a pointer yields the object to which the pointer points.
- States of address value stored in a pointer:
 - point to an object
 - null pointer, not bound to any object: `nullptr`

Remark 1.11 (Initialize all pointers). *Define a pointer only after the object to which it should point, or initialize the pointer to `nullptr`.*

Example 1.12 (Pointers to pointers).

```
int ival = 1;
int *pi = &ival;
int **ppi = &pi;
```

Example 1.13 (References to pointers).

```
int i = 1;
int *p;
int *&r = p;
```

1.4 const

References

- [1] S.B. Lippman, J. Lajoie, and B.E. Moo. *C++ Primer*. Pearson Education, 2012. ISBN: 9780133053036.